

CHAPTER 1 | Introduction

ARMY



ARMY BASE



Why database?

Imagine you're the owner of a fruit store franchise, around 30 years ago. You keep records manually and there are records called

- > sales
- > inventory
- > suppliers



FRUIT SHOP RECORDS

Sales

- Apple [100]
- Banana [80]
- Papaya [90]
- Kiwi [200]
- Guava [50]
- .
- .

Inventory

- Apple [20 kg]
- Banana [40 kg]
- Guava [30 kg]
- Papaya [40 kg]
- Kiwi
- .
- .

Suppliers

- Hari, Banepa,
Apples, 40 kg, 80
- Gopal, Gorkha,
Apples, 20 kg, 75
- Shyam, Dhading,
Guava, 20 kg, 40
- .

Remedies by Database

1. Avoid Data Conflict
2. Mitigation of data redundancy
3. Concurrent Access
4. Backup and Recovery
5. Scalability
6. Search and Retrieval

Therefore, Database can be defined as, “an organized collection of information stored electronically that lets multiple people safely access, manage, and update the same data without conflicts or duplicates.”

Evolution of Database

...

#Assignment 0.1

...

What is Data Abstraction?

ab·stract

['abstrakt]

**existing in thought or as an idea but not
having a physical or concrete existence**

Data Abstraction is a fundamental concept in database management where the complex, low-level details of how data is stored and managed are hidden from users, presenting them instead with a simplified, logical view of the data.

How is Data Abstraction achieved?

Simple answer,

Using the three schema architecture.

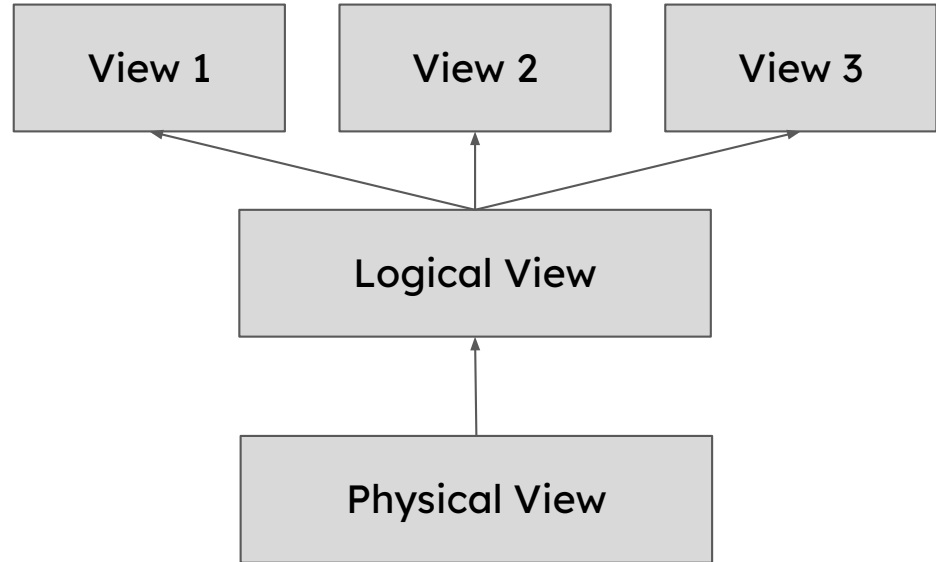
What is the 3 Schema Architecture?

The Three-Schema Architecture is a database framework that implements data abstraction by creating multiple, isolated levels of data representation.

Layers of 3 Schema Architecture

The 3 Schema Architecture is implemented by 3 isolated layers, namely:

- > physical layer
- > logical layer
- > view layer



Data Abstraction

Physical (Internal) Level

Describes how the data is physically stored on the storage media. It is the most complex level.

Viewed by:

DBMS Developers, System
Programmers



Data Abstraction

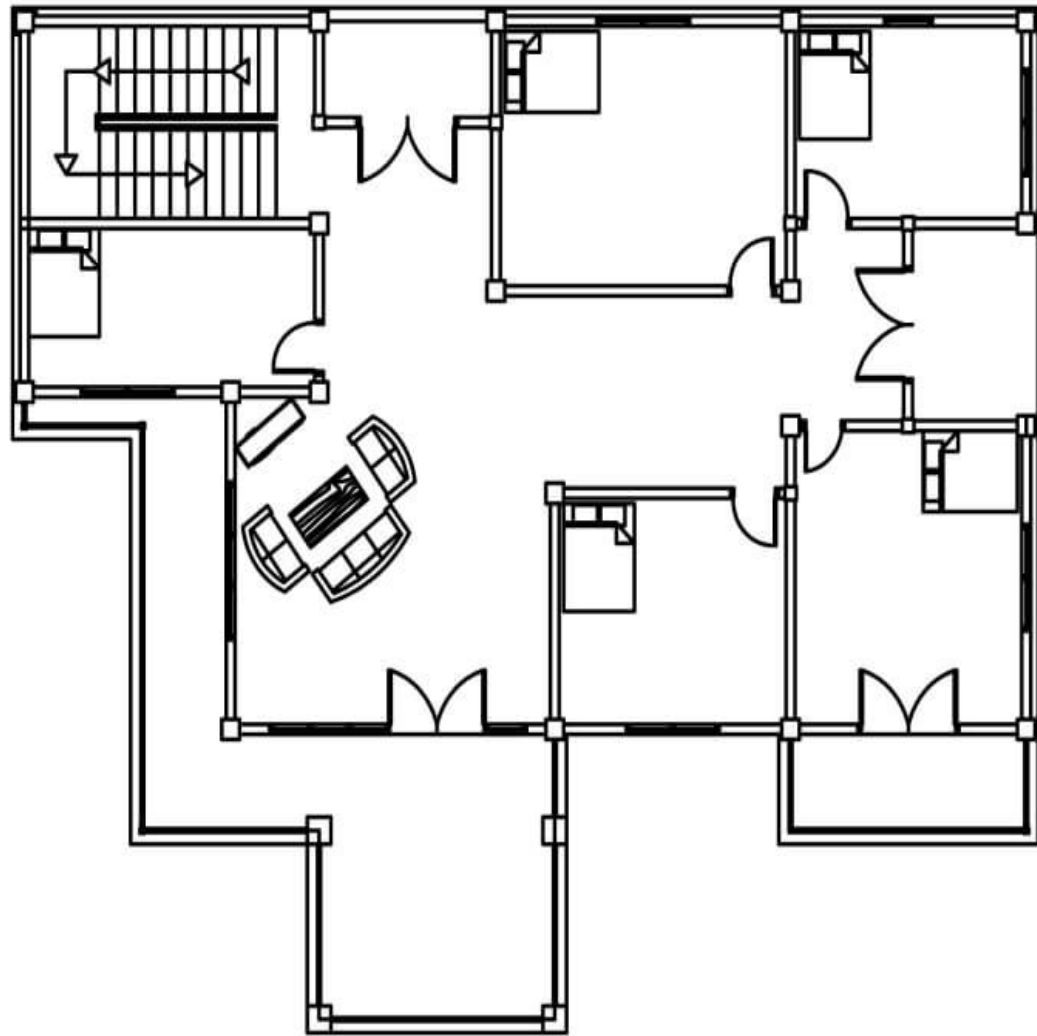
Conceptual (Logical) Level

It describes what data is stored and the relationships among them.

Viewed by:

Database Administrators
(DBAs), Designers

Also called SCHEMA



Data Abstraction

View (External) Level

The user's view of the database. Different users see different, customized views of the data.

Viewed by:

End Users, Application
Programmers

Also called INSTANCE



DATA INDEPENDENCE

It is the ability to modify the database schema at one level without affecting the schema at the next higher level. It's what allows databases to evolve and improve without breaking existing applications.

It is of two types:

1. Physical Data Independence
2. Logical Data Independence

Physical Data Independence

It is the ability to modify the physical storage details without changing the logical tables that applications use.

You can change *HOW* the data is stored without affecting the logical structure or applications.

E.g

- > data compression methods
- > storage location

Logical Data Independence

The ability to modify the logical structure (tables, columns, relationships) without changing the external views that users see.

You can change WHAT data is stored and how it's organized without affecting how users see the data.

E.g

- > add new columns
- > split columns

Significance of Data Abstraction

- Hides complex storage details from users
- Presents a simplified, logical view of data
- Allows physical storage changes without breaking applications
- Enables database structure modifications without affecting users
- Provides different data views for different users
- Protects applications from underlying data changes

Significance of Data Abstraction (*continued*)

- Enhances database security and access control
- Makes systems more manageable and adaptable
- Separates data storage from data usage
- Supports multiple applications using the same database
- Simplifies application development and maintenance
- Provides data independence at multiple levels

What are 'Schema' and 'Instance'?

Schema

A schema is the logical structure that defines the organisation of database objects, their interrelationships and the constraints in the data.

```
CREATE TABLE Products (  
    product_id INT,  
    product_name VARCHAR(50),  
    price DECIMAL(10,2)  
);
```

Instance

An instance is the collection of all actual data stored in the database at a particular point in time.

product_id	product_name	price
101	ABC	123
102	MNO	567
.	.	.
.	.	.

DATABASE vs DBMS

The picture alongside shows a library.

Question:

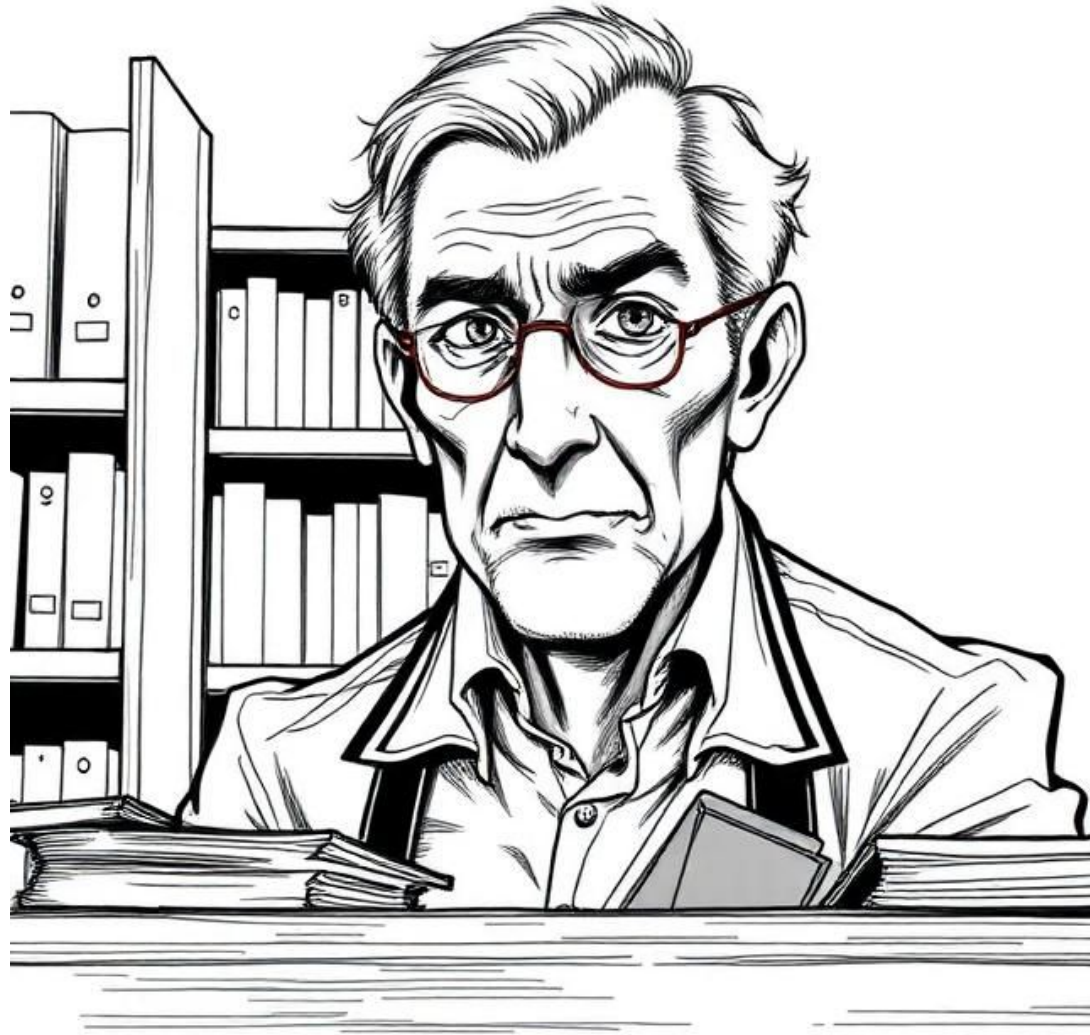
Is the library a Database or a Database Management System?



DATABASE vs DBMS

Assignment #0.2

Write the difference between Database and Database Management Systems.



DATABASE LANGUAGES

Database Languages are specialized programming languages used to manage and manipulate databases. They serve as the interface between users and the database management system (DBMS), allowing for definition, manipulation, and control of data.

Main Types:

- DDL
- DML
- DCL
- TCL

Data Definition Language

DDL (Data Definition Language)

- Defines and modifies the database structure itself
- Creates tables, indexes, views, and other objects
- Establishes the framework that holds data

Commands: CREATE, ALTER, DROP, TRUNCATE

Used by: Database Administrators (DBAs), System Architects

DDL | Create

Creates new database objects (tables, databases, etc.)

DDL | Alter

ALTER statement is used to modify the structure of an existing database object, primarily tables. It allows you to change the table definition without having to drop and recreate the table (which would cause data loss).

```
CREATE TABLE Employee (  
    emp_id INT PRIMARY KEY,  
    name VARCHAR(25),  
    post VARCHAR(25),  
    email VARCHAR(25)  
);
```

DDL | Alter

Use cases:

1. Add single column
2. Add multiple columns
3. Modify single column
4. Modify multiple columns
5. Dropping multiple columns
6. Rename table
7. Rename Column
8. Adding Primary Key
9. Adding Foreign Key, etc.

```
ALTER TABLE Employee
```

```
ADD CONSTRAINT
```

```
    fk_employee_dept
```

```
FOREIGN KEY (department_id)
```

```
REFERENCES
```

```
    Department(dept_id);
```


DDL | Drop

The DROP statement in MySQL is used to completely remove database objects.

Unlike ALTER which modifies, DROP deletes objects permanently.



Warning: Irreversible!

```
-- Drop single table
```

```
DROP TABLE Employee;
```

```
-- Drop multiple tables
```

```
DROP TABLE Employee,  
Department, Project;
```

DDL | Truncate

TRUNCATE is specifically for removing all rows from a table while keeping the table structure intact.

```
-- Remove all rows
```

```
TRUNCATE TABLE Employee;
```

```
-- Truncate multiple tables
```

```
TRUNCATE TABLE Employee,  
Department, Project;
```

Data Manipulation Language

DML (Data Manipulation Language)

- Handles the actual data within the structures
- Used for daily operations: adding, reading, updating data
- Includes both simple and complex data queries

Commands: INSERT, SELECT, UPDATE, DELETE

Used by: Application Developers, End Users, Data Analysts

DML | Insert

INSERT is the SQL statement used to add new rows of data into database tables.

It's part of the C in CRUD (Create).

```
-- Single row with all  
columns
```

```
INSERT INTO employee VALUES  
(1, 'John', 'john@email.com',  
'2024-01-01');
```

```
-- Multiple rows in one  
statement
```

```
INSERT INTO employee (name,  
email) VALUES  
( 'Bob', 'bob@email.com'),  
( 'Carol', 'carol@email.com'),  
( 'Dave', 'dave@email.com');
```

DML | Select

SELECT statement performs read operations on databases, returning result sets that match specified criteria without modifying the underlying data.

-- Select all columns

SELECT * FROM employee;

-- Select specific columns

SELECT id, name, email

FROM employee

WHERE salary>10000;

DML | Update

UPDATE is the SQL statement used to modify existing data in database tables.

It's part of the U in CRUD (Update).

```
-- single record update
```

```
UPDATE users SET status =  
'active' WHERE id = 123;
```

```
-- multiple record update
```

```
UPDATE products
```

```
SET
```

```
    price = price * 0.9,
```

```
    discount_applied = TRUE
```

```
WHERE category = 'Clearance';
```

DML | Delete

DELETE is the SQL statement used to remove rows from database tables.

It's part of the D in CRUD (Delete).

```
-- Select all columns
```

```
DELETE FROM employee;
```

```
-- Select specific columns
```

```
DELETE FROM employee
```

```
WHERE salary>10000;
```

Data Control Language

DCL (Data Control Language)

- Controls access rights and permissions
- Determines who can access what data and how
- Essential for database security and privacy

Commands: GRANT, REVOKE

Used by: Database Administrators, Security Officers

Transaction Control Language

TCL (Transaction Control Language)

- Ensures data integrity during operations
- Manages transaction sequences and recovery
- Maintains database consistency

Commands: COMMIT, ROLLBACK, SAVEPOINT

Used by: Application Developers, Database Administrators

Assignment #1

1. Define the terms [Database, DBMS, Schema, Instance, Database Languages, File-processing systems]
2. List four significant differences between file-processing systems and a DBMS.
3. Describe the different levels of data abstraction. Discuss the significance of this abstraction.
4. Explain the concept of physical data independence and its importance in database systems.
5. What are the 5 main functions of Database Programmer and Database Architects.